

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
8	(a)	(i)		{kill/prevent growth} of bacteria (in wounds)	1			1		
		(ii)		virus	1			1		
		(iii)		possible side effects/ unknown long term effects	1			1		
	(b)			Any one (x1) from pH + {add/use} acid/alkali reservoir oxygen + sterile air in temperature + water jacket/ cold water in		1		1		1
	(c)	(i)		0.5 = 2 marks 15/30 or (23-8)/30 = 1 mark		2		2	2	
		(ii)	I	Accept any figure between 96-100 <u>hours</u> (1) Mass of penicillin is at its {maximum/ remains constant/ levels off}/ no more penicillin is being produced/ penicillin production has stopped/ OWTTE (1)		1	1	2		
	(d)			Needs to start at same point, end same level but to left of existing curve for Penicillium			1	1		
				Question 8 total	3	4	2	9	2	1